

Tilt-Based Control Barrier Function for Target Visibility

A camera-plane QP over the body tilt via the rotational interaction matrix

Soft-Precise-Landing

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Scope. Visibility is an *attitude* constraint — the body tilt moves the field of view — so we guarantee it with a small QP (a clamp) on the desired tilt, on the *real* camera plane. The coupling between tilt and feature is the **rotational part of the IBVS interaction matrix**, L_ω , evaluated at the measured image point: it is exact, *depth-independent*, and needs no de-rotation / virtual frame. The barrier uses image features only — scale- and depth-free (no β , no optic flow, no disturbance term). Notation follows the manuscript.

Notation

Symbol	Meaning
${}^c\hat{\mathbf{r}}_i, {}^c\hat{\mathbf{r}} = (x, y)$	camera image feature points / centroid, tangent units, per axis k
$L_\omega({}^c\hat{\mathbf{r}})$	rotational interaction matrix (roll/pitch columns), $\partial {}^c\dot{\hat{\mathbf{r}}} / \partial \boldsymbol{\omega}_{\text{rp}}$
$\boldsymbol{\omega}_{\text{rp}} = [\omega_\phi, \omega_\theta]$	body roll/pitch rate (the tilt rate, the control)
$\boldsymbol{\theta}, \theta_d, \theta^*$	body tilt per axis; desired tilt (from R_d), filtered
$\mathbf{d}$	exogenous (translation) feature drift, $\mathbf{d} = {}^c\dot{\hat{\mathbf{r}}}_{\text{obs}} - L_\omega \boldsymbol{\omega}_{\text{rp}}^{\text{curr}}$
$\mathcal{R}, f$	resolution, focal (px); $\varphi_{\text{max}} = \mathcal{R}/(2f)$ half-FoV (tangent), the FoV edge
$\delta, \dot{\delta}$	primary-marker half-spread (loom rate)
θ_{cap}	deliverable tilt cap (60°); τ drift look-ahead; h barrier

1 Barrier on the real camera plane

Per image axis k ,

$$h_k = \varphi_{\text{max},k} - |{}^c\hat{\mathbf{r}}_k| - \delta_k. \quad (1)$$

$h_k \geq 0 \iff$ the marker (outermost feature point) is on the sensor. ${}^c\hat{\mathbf{r}}$ is the *measured* camera image point — nothing is de-rotated.

2 Tilt→feature coupling: the rotational interaction matrix

A camera rotation moves a feature point by the rotational part of the IBVS interaction matrix, a standard and **depth-independent** object evaluated at the measured point ${}^C\hat{\mathbf{r}} = (x, y)$:

$${}^C\dot{\hat{\mathbf{r}}} = L_\omega({}^C\hat{\mathbf{r}})\boldsymbol{\omega}_{\text{rp}} + \mathbf{d}, \quad L_\omega = \begin{bmatrix} xy & -(1+x^2) \\ 1+y^2 & -xy \end{bmatrix}, \quad (2)$$

where $\boldsymbol{\omega}_{\text{rp}}$ is the body roll/pitch rate (the control) and $\mathbf{d}$ is the exogenous (translation / target) drift. L_ω contains **no depth** — the tilt coupling needs no β and no virtual frame. The drift is measured directly by stripping the rotational term from the observed feature velocity, $\mathbf{d} = {}^C\dot{\hat{\mathbf{r}}}_{\text{obs}} - L_\omega\boldsymbol{\omega}_{\text{rp}}^{\text{curr}}$, so it carries only the genuine target motion, not our own tilting.

For a finite tilt change the exact map is the rotation homography of the measured point; the first-order form ${}^C\hat{\mathbf{r}}(\boldsymbol{\theta}) \approx {}^C\hat{\mathbf{r}} + L_\omega(\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{curr}})$ is the local linearization, accurate over one control step.

3 The visibility filter: a QP over the body tilt

With $m_k = \varphi_{\text{max},k} - \delta_k - \tau\dot{\delta}_k$ the predicted margin,

$$\boxed{\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} \|\boldsymbol{\theta} - \boldsymbol{\theta}_d\|^2 \text{ s.t. } |{}^C\hat{\mathbf{r}}_k + [L_\omega(\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{curr}})]_k + \tau d_k| \leq m_k, \quad |\boldsymbol{\theta}| \leq \theta_{\text{cap}}} \quad (3)$$

A small 2-variable QP (the two tilt axes), with ≤ 2 binding rows plus the cap — closed-form. L_ω is near-diagonal for a centred marker (the xy off-terms vanish), so it is nearly a per-axis clamp; the off-diagonal carries the genuine roll↔pitch cross-coupling near the FoV edge.

- **Directional** — only the outward binding row is active; inward (recovery) and tangential tilts are free. No strangling.
- **Input-aware** — the θ_{cap} box never demands a tilt the inner loop cannot hold.
- **Forward-invariant** — the look-ahead uses the predicted margin in τ s: the target drift $\mathbf{d}$ (left) and the marker-fill rate $\dot{\delta}$ (right) shrink it, so the box tightens early and forces the inward tilt ($\dot{h} \geq -\alpha(h)$). $\tau = 0$ gives the static clamp.

4 Two-phase δ (central-marker convention)

The designated *central* marker is the target. **Phase 1 — while it decodes**, $\delta_k = 0$: the barrier is centroid-only, keeping the target centre in frame while deliberately letting the central marker grow and overflow as the UAV closes on it (its visibility *is* the guarantee; using its large extent would fire the barrier early). **Phase 2 — on its overflow** (decode-fail, hysteresis-gated against flicker $\Rightarrow$ touchdown range), δ **ramps in** over a few frames from the visible-*set* half-extent ($\frac{1}{2}\text{ptp}_k$ over ALL visible markers' corners, so $|{}^C\hat{\mathbf{r}}| + \delta$ is the true outermost feature — not one marker) and tracks them thereafter. The ramp (not a step) keeps h from dropping discontinuously at engagement, but clamp the ramp gain so δ never *lags* the measured $\frac{1}{2}\text{ptp}$ — a ramp slower than the fill under-protects. Within Phase 2 the visible markers themselves leave (a mild inner sawtooth); $\dot{\delta}$ is reset across those switches.

5 Feasibility floor and hand-off

The QP (3) is **infeasible** iff the predicted margin is gone ($m_k < 0$ — the marker fills the frame within τ) or even the full $L_\omega \theta_{\text{cap}}$ tilt cannot bring ${}^C\hat{\mathbf{r}}$ inside. Then visibility is physically impossible $\Rightarrow$ raise the perception-floor flag and hand off to the texture-free ring flow.

6 Assumptions (and the practical challenge each carries)

1. **L_ω 's sign / axis convention is calibrated.** Its *form* is the standard interaction matrix, but the body-rate $\rightarrow$ image-axis sign (roll $\rightarrow$ which axis, which direction) depends on the camera mount. *Must-do:* command a small tilt, check $\Delta {}^C\hat{\mathbf{r}}$ matches L_ω in sign, before trusting the barrier.
2. **The filtered tilt is realized** (SO(3) loop tracks $\boldsymbol{\theta}^*$). *Challenge:* attitude bandwidth $\Rightarrow$ lag; θ_{cap} bounds magnitude, not delay.
3. **The drift $\mathbf{d}$ is the translation part** (${}^C\dot{\hat{\mathbf{r}}}_{\text{obs}} - L_\omega \boldsymbol{\omega}^{\text{curr}}$) and $\dot{\delta}$ the loom. *Challenge:* finite differences $\Rightarrow$ filter + binding-axis hysteresis; the disturbance enters as *measured* motion — no separate $\mathbf{d}_h$ term. The homography centroid (hence ${}^C\dot{\hat{\mathbf{r}}}_{\text{obs}}$) must stay *smooth across marker-set changes* (the homography re-fits on a different visible subset) or $\mathbf{d}$ spikes — filter / hold a stable subset across a switch.
4. **First-order L_ω over the cycle.** Exact is the rotation homography of ${}^C\hat{\mathbf{r}}$; the $L_\omega(\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{curr}})$ linearization is accurate over one control step. At $\theta_{\text{cap}} = 60^\circ$ the integral (homography) is the safe choice for the boundary $\boldsymbol{\theta}$.
5. **θ_{cap} is the true deliverable tilt** (thrust unsaturated).

7 Scale-freeness and depth-freeness

The barrier (1), the coupling (2), and the box (3) contain only the measured image point ${}^C\hat{\mathbf{r}}$, δ , $\varphi_{\text{max}} = \mathcal{R}/(2f)$, the body rate, and the tilt. L_ω is **depth-independent by construction** (no $\mathcal{V}z_t$, no β), so the whole formulation is manifestly scale- and depth-free. The only metric quantity upstream is the geometric tracker's R_d (a controller output).

8 Implementation (per control cycle)

1. (*once*) Calibrate L_ω 's sign / axis convention (Assumption 1).
2. Measure ${}^C\hat{\mathbf{r}}$ (board centroid, homography). Two-phase δ : 0 while the central marker decodes; on overflow (hysteresis) ramp in the visible markers' $\frac{1}{2}$ ptp.
3. Evaluate $L_\omega({}^C\hat{\mathbf{r}})$; form the drift $\mathbf{d} = {}^C\dot{\hat{\mathbf{r}}}_{\text{obs}} - L_\omega \boldsymbol{\omega}^{\text{curr}}$ and $\dot{\delta}$ (filtered).
4. Predicted margin $m = \varphi_{\text{max}} - \delta - \tau\dot{\delta}$; build the QP (3).
5. $\boldsymbol{\theta}_d$ from R_d (equivalently the desired lateral accel via $\tan \theta_d = \mathcal{I}_{a_d,xy}/a_z$); solve $\rightarrow \boldsymbol{\theta}^*$ (a 2-var clamp).

6. If infeasible, raise the perception-floor flag and switch the flow source to the ring estimator;
else pass $\boldsymbol{\theta}^*$ (equivalently the clamped a_{xy}) to the SO(3) inner loop.